

PhD Proposal: Auditable Artificial Intelligence for ESG Intelligence: Explainable Linguistic Analysis, Neurosymbolic Reasoning, and Knowledge-Driven Verification

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Abstract

Environmental, Social, and Governance (ESG) disclosures have become a fundamental determinant of corporate creditworthiness, yet their effectiveness is hampered by a persistent *discourse-practice gap* (Brogi et al., 2022; Santos et al., 2025). In this gap, corporate narratives are decoupled from substantive environmental actions, often manifesting as *carbonwashing*—a strategic misrepresentation of emission reductions that carries profound regulatory and financial weight (In and Schumacher, 2021; Santos et al., 2025). Empirical evidence suggests that symbolic compliance is widespread; for instance, a 1% increase in an entity’s AI-calculated environmental score can paradoxically correlate with a 5.24% increase in actual greenhouse gas emissions (Santos et al., 2025). This doctoral research proposes an auditable AI framework to unmask such deceptive rhetoric by integrating domain-adapted Natural Language Processing (NLP), neurosymbolic reasoning, and multimodal data fusion. The methodology leverages FinBERT, a domain-adapted model capable of identifying 18% more textual informativeness than generic architectures by capturing “abnormal positive tone”—a common marker of managerial attempts to obfuscate poor sustainability performance (Huang et al., 2022; Xin et al., 2024). The framework consists of three pillars: (i) Explainable Aspect-Based Sentiment Analysis (ABSA) to visualize linguistic features triggering greenwashing signals; (ii) a neurosymbolic architecture that embeds formal regulatory logic (e.g., EU Taxonomy) into neural pipelines to provide transparent decision traces; and (iii) Knowledge Graph-extended Retrieval-Augmented Generation (KG-RAG) to ground assessments in verifiable multimodal evidence (text, tables, and images) (Yim et al., 2025; Kaoukis et al., 2025). By implementing self-correction protocols like Chain-of-Verification (CoVe), the system reduces hallucination rates from 16% to 10%, transforming opaque “black box” ratings into auditable “open box” intelligence (Yim et al., 2025). This research provides a scalable solution for both Multinational Corporations (MNCs) and resource-constrained Small and Medium Enterprises (SMEs), ensuring accountability in the global sustainable finance ecosystem (Chou, 2024).

Keywords

Aspect-Based Sentiment Analysis, Explainable AI, Carbonwashing Detection, ESG Analytics, Neurosymbolic Reasoning, Knowledge Graphs, Retrieval-Augmented Generation, Chain-of-Verification, Sustainable Finance, Regulatory Compliance

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Opening Summaries

Project Summary

This project develops an **auditable AI architecture for ESG intelligence** designed to resolve "aggregate confusion"—the systemic divergence in ESG ratings caused by methodological opacity and subjective qualitative weights (Censi et al., 2025; Vitto et al., 2023). The research shifts ESG assessment from static, annual reports, which suffer from a significant "temporal lag" and 90% omission rates of negative events, toward dynamic, stream-based intelligence systems (Jeong et al., 2025; Boiral, 2013). The novelty of the proposed framework lies in three integrated layers:

1. **Explainable Detection:** Utilizing FinBERT-ABSA and XAI tools (SHAP/LIME) to identify "strategic fluffing" and "impression management" at a granular, sentence-level scale (Prajwalasimha et al., 2025).
2. **Regulatory Reasoning:** A neurosymbolic engine that enforces logical consistency with established sustainability taxonomies, ensuring that neural predictions do not violate legal guardrails (Heever et al., 2024; Wu and Zhao, 2025).
3. **Multimodal Verification:** A KG-RAG pipeline that cross-references corporate claims against objective performance metrics and external signals (news, litigation) to identify tone–evidence mismatches (Santos et al., 2025; Nassar et al., 2025).

The resulting system provides a "justification-centric" assessment, bridging the data scarcity gap for SMEs through bridging strategies adapted from large-cap benchmarks (Chou, 2024; Prajwalasimha et al., 2025).

Non-Technical Summary

As climate change reshapes financial markets, companies use sustainability reports to prove their environmental commitment. However, many firms engage in "greenwashing" or "carbonwashing"—using overly optimistic language to hide a lack of real progress (In and Schumacher, 2021; Sisdyani et al., 2025). Because these reports are long and complex, it is difficult for investors and regulators to tell the difference between genuine action and "strategic fluffing" (Pope et al., 2023). This research creates an intelligent "open box" system that reads these reports like a human auditor but with the speed and precision of advanced AI. Unlike current systems that give a single, unexplained score, this tool shows its work. It identifies misleading phrases, connects them to official environmental rules (like the EU Taxonomy), and cross-checks them against real data, such as actual carbon emission records (Yim et al., 2025; Santos et al., 2025). Importantly, this system is designed to help Small and Medium Enterprises (SMEs), which often struggle with the high costs of complex reporting (Kassem and Trenz, 2020). By using automated verification and self-correction tools, the system reduces errors and "hallucinations," providing a trustworthy way to ensure that corporate promises match environmental reality (Peer and Stabinger, 2025). The ultimate goal is to restore trust in green investing and ensure that capital flows to companies making a real impact on the planet.

1 Research Context and Motivation

1.1 Problem Statement

Environmental, Social, and Governance (ESG) performance is increasingly recognized as a fundamental determinant of corporate creditworthiness, with strong ESG awareness scores associated with reduced credit risk across diverse industries (Brogi et al., 2022; Kambe and Tamamura, 2022b). However, the effectiveness of ESG-driven capital allocation is fundamentally undermined by a persistent *discourse–practice gap*, where symbolic corporate narratives are decoupled from substantive environmental actions (Santos et al., 2025; In and Schumacher, 2021). This phenomenon manifests most severely as *carbonwashing*—the selective misrepresentation of progress in carbon emission reductions, which distorts market signals in the post-Paris Agreement landscape (In and Schumacher, 2021).

The detection of such deceptive rhetoric is complicated by the "black box" nature of current assessment models, which rely on proprietary, non-transparent methodologies that produce inconsistent ratings—a state termed *aggregate confusion* (Censi et al., 2025; Vitto et al., 2023; Giri and Chaparro, 2024). Current Natural Language Processing (NLP) pipelines often exacerbate this confusion; generic models frequently mislabel specialized financial language as neutral, failing to capture the "abnormal positive tone" and "strategic fluffing" used by management to obfuscate poor sustainability performance (Huang et al., 2022; Pope et al., 2023). Consequently, there is an urgent need for an auditable intelligence framework that unifies linguistic clarity with objective evidence while maintaining regulatory alignment.

1.2 Evidence of Importance and Urgency

The urgency of this research is driven by a reliability crisis in corporate disclosures and the structural exclusion of resource-constrained entities from the sustainable finance ecosystem.

Reliability Crisis and Data Mismatches Empirical evidence suggests that corporate reporting often functions as a "simulacrum" of responsibility; up to 90% of significant negative environmental events are omitted from official reports (Boiral, 2013; Zhu et al., 2025). This decoupling results in paradoxical correlations where a 1% increase in an entity's AI-calculated environmental score can be associated with a 5.24% increase in actual greenhouse gas emissions (Santos et al., 2025). Without mechanisms to cross-validate claims against physical performance data, ESG ratings risk rewarding image management over tangible stewardship.

The SME Representation Gap Small and Medium-sized Enterprises (SMEs) face a significant "representation gap" in the sustainable finance ecosystem (Sandanayake et al., 2023; Chou, 2024). Unlike Multinational Enterprises (MNEs), SMEs lack the financial resources and technical expertise to navigate exhaustive reporting frameworks like the CSRD, viewing ESG reporting as a "superfluous and burdensome" activity (Kassem and Trenz, 2020). Current evaluation models, designed for large-cap firms, operate as a "privilege system," effectively excluding SMEs from green investment opportunities due to data scarcity and high collection costs (Alkatheeri et al., 2023; Chou, 2024).

Temporal Lag in Static Reporting Traditional ESG assessments rely on annual, static reports that are often months or years out of date by the time they are processed (Jeong et al., 2025; Nassar et al., 2025). This "temporal lag" leaves stakeholders blind to rapidly evolving risks, such as sudden environmental disasters or labor strikes (Saxena et al., 2022). Transitioning toward dynamic, stream-based intelligence using real-time news and social media signals is critical for capturing an immediate "photograph" of a firm's risk profile (Nassar et al., 2025; Jaiswal et al., 2024).

1.3 Research Gap and Justification

Despite advancements in domain-adapted models like FinBERT, four critical gaps persist in the development of auditable ESG intelligence:

1. **Explainability Gap (RQ1):** Current models prioritize coarse sentiment polarity rather than fine-grained, aspect-level explanations. There is a lack of integration between Aspect-Based Sentiment Analysis (ABSA) and XAI tools (e.g., SHAP, LIME) to visualize the linguistic features that trigger greenwashing signals (Huang et al., 2022; Černevičienė and Kabašinskas, 2024).
2. **Regulatory Alignment Gap (RQ2):** High-performing deep learning models lack the logical rigor of symbolic reasoning. Existing systems cannot enforce "built-in safety guardrails" to ensure scores remain consistent with formal taxonomies like the EU Taxonomy or GRI (Heever et al., 2024; Wu and Zhao, 2025).
3. **Multimodal Verification Gap (RQ3):** Most frameworks are unimodal, ignoring the dense quantitative data found in tables, layouts, and external reputation indices. There is an unresolved challenge in utilizing Knowledge Graph-extended Retrieval-Augmented Generation (KG-RAG) to ground LLM evaluations in verifiable multimodal evidence across both MNEs and SMEs (Zhang et al., 2025; Kaoukis et al., 2025; Linders and Tomczak, 2025).
4. **Factual Integrity Gap (RQ4):** The deployment of Large Language Models (LLMs) in high-stakes regulatory environments is hampered by factual hallucinations and overconfidence. Current systems lack automated self-correction protocols, such as Chain-of-Verification (CoVe), to ensure generated disclosures are robust against "automated greenwashing" (Wu et al., 2025; Linders and Tomczak, 2025; Peer and Stabinger, 2025).

This research justifies the integration of neurosymbolic architectures and KG-RAG as a pathway toward "open box" ESG intelligence. By anchoring corporate claims to sector-specific materiality and verifiable physical outcomes, the proposed framework transforms subjective narratives into a structured, auditable metric suitable for modern sustainable finance (Yim et al., 2025; Censi et al., 2025; Santos et al., 2025).

2 Research Questions, Objectives, and Hypotheses

2.1 Research Questions

This doctoral research investigates how explainable, neurosymbolic, and knowledge-driven artificial intelligence can transform ESG analytics from opaque prediction systems into auditable, regulator-aligned decision support infrastructures. The research is structured around four interrelated research questions (RQs), progressing from micro-level linguistic interpretability to system-level regulatory intelligence and robustness.

RQ1 — Explainable Linguistic Detection of Impression Management *How can Aspect-Based Sentiment Analysis (ABSA), when integrated with Explainable AI techniques such as SHAP and LIME, enable reliable identification of impression management strategies—specifically carbonwashing and strategic fluffing—by attributing model predictions to concrete linguistic features in sustainability disclosures?*

This question focuses on sentence- and aspect-level interpretability, examining whether explainable ABSA models can expose the linguistic mechanisms through which misleading sustainability narratives are constructed and detected.

RQ2 — Neurosymbolic Trade-off Between Predictive Accuracy and Interpretability *To what extent can neurosymbolic architectures reconcile the trade-off between predictive performance and interpretability in ESG risk models by embedding symbolic regulatory logic (e.g., EU Taxonomy and CSRD constraints) within neural learning pipelines to generate transparent decision traces suitable for regulatory audit?*

This question investigates whether hybrid reasoning systems can preserve high predictive utility while delivering formally grounded explanations aligned with regulatory semantics.

RQ3 — Multimodal Knowledge-Graph-Extended Retrieval for Justification-Centric Intelligence *How can Knowledge Graph-extended Retrieval-Augmented Generation (KG-RAG) integrate multimodal evidence (text, tables, and visual layouts) to support scalable, justification-centric ESG intelligence capable of real-time verification and fact-checking across heterogeneous organizational contexts?*

This question evaluates the role of structured knowledge and multimodal grounding in producing verifiable, auditable, and scalable sustainability intelligence.

RQ4 — Robustness and Self-Correction in Generative ESG Systems *How can generative reporting and verification systems integrate automated self-correction mechanisms—such as Chain-of-Verification and XAI-enabled compliance validation—to reduce hallucination risk and prevent automated greenwashing in high-stakes regulatory environments?*

This question addresses system reliability, safety, and governance under real-world deployment constraints.

2.2 Research Objectives

The overarching goal of this research is to design, implement, and evaluate an auditable AI framework for ESG intelligence that delivers transparent, evidence-grounded, and regulator-aligned decision support.

The specific objectives are:

1. **Design explainable ABSA pipelines** capable of extracting fine-grained aspect-sentiment structures and attributing predictions to interpretable linguistic features associated with impression management.
2. **Develop neurosymbolic ESG reasoning modules** that encode regulatory logic into neural models to enable traceable, constraint-consistent decision pathways.

3. **Construct a multimodal KG-RAG architecture** that fuses textual, tabular, and layout-aware evidence into justification-centric ESG assessments.
4. **Integrate automated self-verification mechanisms** (e.g., Chain-of-Verification, alignment validation) to improve factual robustness and reduce hallucination risk.
5. **Evaluate predictive performance, interpretability, and auditability** across real-world ESG datasets using quantitative metrics and qualitative audit simulations.
6. **Demonstrate scalability and accessibility** for both multinational enterprises and resource-constrained SMEs through computational efficiency analysis and deployment-oriented benchmarking.

2.3 Hypotheses

The research will empirically test the following hypotheses:

- **H1 (Explainability Gain):** Explainable ABSA models augmented with SHAP or LIME will achieve significantly higher interpretability fidelity and error localization accuracy than non-explainable baselines while maintaining comparable classification performance.
- **H2 (Neurosymbolic Transparency):** Neurosymbolic ESG models embedding regulatory constraints will reduce explanation inconsistency and regulatory violation rates without materially degrading predictive accuracy compared to purely neural architectures.
- **H3 (Multimodal Justification Quality):** KG-RAG systems integrating multimodal evidence will produce more verifiable and auditor-consistent justifications than text-only retrieval or end-to-end generative baselines.
- **H4 (Robustness via Self-Correction):** Systems incorporating Chain-of-Verification and compliance validation will exhibit statistically significant reductions in hallucination frequency and contradiction rates under adversarial or noisy inputs.
- **H5 (Scalability):** The proposed architecture will demonstrate acceptable latency, memory usage, and annotation efficiency suitable for deployment in SME-scale environments without loss of analytical reliability.

2.4 RQ–Objective–Metric Alignment

RQ	Primary Objective	Evaluation Metrics
RQ1	Explainable ABSA for impression management detection	F1, Precision/Recall, SHAP faithfulness, explanation stability, human agreement
RQ2	Neurosymbolic regulatory reasoning	Accuracy, constraint violation rate, explanation consistency, trace completeness
RQ3	Multimodal KG-RAG justification	Evidence coverage, citation accuracy, audit agreement, retrieval recall
RQ4	Robustness and self-correction	Hallucination rate, contradiction rate, verification success rate

3 Theoretical Framework and Literature Positioning

3.1 Conceptual Foundations

This research is grounded in a multidisciplinary theoretical synthesis spanning sustainability accounting, computational linguistics, and hybrid artificial intelligence (AI). Environmental, Social, and Governance (ESG) performance is increasingly recognized as a fundamental determinant of corporate creditworthiness, with strong ESG awareness scores associated with a reduction in firm credit risk (Brogi et al., 2022; Kambe and Tamamura, 2022a). However, the effectiveness of these metrics is undermined by the "discourse–practice gap," where internal corporate narratives are decoupled from objective performance metrics (Santos et al., 2025). The framework is anchored by three conceptual pillars: organizational legitimacy via symbolic compliance, human-centric explainability, and neurosymbolic reasoning.

3.2 Discourse–Practice Gap and Organizational Legitimacy

The core theoretical lens for investigating "carbonwashing" is rooted in the "discourse–practice gap," where a firm's symbolic sustainability claims are decoupled from its substantive environmental actions (In and Schumacher, 2021). This phenomenon is driven by organizational legitimacy theory, suggesting that corporations utilize symbolic maneuvers to maintain social acceptance while avoiding the high costs of genuine decarbonization. In this environment, firms often resort to "symbolic compliance" rather than substantive action (Santos et al., 2025).

Empirical evidence of this decoupling is found in the "tone–evidence mismatch," where a 1% increase in an entity's AI-calculated environmental score can paradoxically correlate with an approximate 5.24% increase in actual greenhouse gas (GHG) emissions (Santos et al., 2025). Managers utilize "strategic fluffing" and "abnormal positive tone" in annual and sustainability reports to manage stakeholder perceptions (Pope et al., 2023). Traditional sentiment analysis often fails to detect these manipulative frames because generic models mislabel specialized financial language as neutral. Consequently, this research adopts FinBERT, which incorporates finance-specific knowledge to identify approximately 18% more textual informativeness than generic architectures by capturing the linguistic markers of "impression management"

3.3 Explainable AI and Human-Centric Accountability

Explainable AI (XAI) addresses the "black box" nature of current ESG scoring models, which rely on opaque deep learning architectures that lack human-readable justifications. This transparency gap leads to "aggregate confusion," where divergent methodologies and subjective interpretations of qualitative data produce inconsistent ratings for the same entity (Censi et al., 2025; Vitto et al., 2023).

To restore institutional trust, this framework utilizes post-hoc explanation methods, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), to provide "justification-centric" assessments (Prajwalasimha et al., 2025). These specialized XAI tools quantify feature importance and provide granular, sentence-level insights into corporate narratives. By transforming "black box" predictions into a transparent "open box" system, the framework ensures that every compliance decision is justifiable and interpretable by human overseers.

3.4 Neurosymbolic Reasoning and Regulatory Constraints

Neurosymbolic AI provides a hybrid paradigm that integrates the pattern recognition strengths of deep neural networks with the structured reasoning of symbolic AI. This architecture addresses the trade-off between predictive accuracy and interpretability by linking neural-derived features to interpretable symbolic concepts

Symbolic components, such as the ESGSenticNet knowledge base, allow for specialized concept parsing while enforcing symbolic constraints grounded in established sustainability taxonomies like the EU Taxonomy

or the Greenhouse Gas (GHG) Protocol (Heever et al., 2024; Pakina and Pujari, 2024). These hybrid systems establish a "unified cognitive loop" that enforces domain constraints and prevents automated systems from violating legal or environmental protocols (Wu and Zhao, 2025). This neurosymbolic approach unmask the gap between rhetoric and practice by cross-referencing neural sentiment analysis with symbolic representations of verifiable corporate actions.

3.5 Knowledge Graphs and Multimodal Justification

Knowledge Graph-extended Retrieval-Augmented Generation (KG-RAG) enables justification-centric intelligence by grounding Large Language Model (LLM) outputs in verifiable, domain-specific evidence (Kaoukis et al., 2025). Knowledge graphs (e.g., EmeraldGraph) surface verifiable evidence that generic LLMs lack, facilitating traversal-based explanations that allow auditors to navigate the lineage of a specific ESG metric down to sentence-level evidence (Yim et al., 2025).

For resource-constrained Small and Medium-sized Enterprises (SMEs), KG-RAG supports "bridging strategies" that utilize causal inference to identify the most relevant sustainability variables from large-cap firms and apply them to data-scarce smaller entities (Chou, 2024). To ensure factual reliability, the system implements the Chain-of-Verification (CoVe) protocol, which has been shown to reduce hallucination rates in high-stakes regulatory assessments. This multimodal verification transforms unstructured "symbolic" narratives into a structured repository of evidence-backed claims.

3.6 Ontological and Epistemological Positioning

Ontologically, this research adopts a *critical realist* stance, positing that while ESG phenomena like actual emissions exist independently of discourse, their interpretation is mediated through institutional narratives that often function as a "simulacrum" of responsibility (Boiral, 2013; Santos et al., 2025). Language is treated as a strategic proxy that may diverge from the physical environmental profile of the firm.

Epistemologically, the research follows a *design science* paradigm, focusing on the construction and evaluation of auditable AI "artefacts" that replicate human auditor judgments. The shift from static, annual reports toward continuous, real-time auditing via stream-based NLP is viewed as a necessary evolution to eliminate the "temporal lag" and subjectivity inherent in traditional disclosure cycles (Jeong et al., 2025; Nassar et al., 2025).

4 Methodology and Research Design

The proposed research architecture transitions from unimodal textual analysis toward an integrated multimodal, neurosymbolic framework designed to resolve "aggregate confusion" and unmask the "discourse-practice gap" in sustainability reporting (Censi et al., 2025; Santos et al., 2025; Chou, 2024). The methodology is structured into six modular layers to ensure transparency, regulatory alignment, and scalability for resource-constrained entities (Huang et al., 2022).

4.1 Research Architecture and Data Triangulation

The system adopts a layered architecture to transform raw disclosures into auditable intelligence (Yim et al., 2025). A critical component is the **triangulation-based detection logic**, which cross-references three distinct data streams to validate corporate claims (Santos et al., 2025):

- **Internal Discourse:** Quantified corporate narratives extracted from PDF reports via domain-specific NLP (Huang et al., 2022).
- **External Signals:** Media reputation indices and news-based sentiment to capture stakeholder perceptions (Yu et al., 2024; Nassar et al., 2025).
- **Objective Metrics:** Physical performance data, specifically greenhouse gas (GHG) emissions, used as a ground-truth validator (In and Schumacher, 2021; Santos et al., 2025).

4.2 Linguistic Intelligence via FinBERT and ABSA

Linguistic extraction utilizes **FinBERT**, a domain-adapted transformer model pre-trained on financial corpora to capture nuances that generic models frequently mislabel as neutral (Huang et al., 2022; Webersinke et al., 2021). This layer implements Aspect-Based Sentiment Analysis (ABSA) to isolate specific sustainability themes (Zhang et al., 2022).

FinBERT's tone analysis is optimized to detect "**abnormal positive tone**"—a linguistic marker of "strategic fluffing" where managers use overly optimistic language to obfuscate poor environmental performance (Xin et al., 2024; Pope et al., 2023). Empirical benchmarks indicate that this domain-specific approach identifies approximately 18% more textual informativeness compared to non-adapted models (Huang et al., 2022).

4.3 Explainability and Aspect Clustering Pipeline

To improve interpretability, the system implements an aspect clustering pipeline that aggregates fine-grained data points into cohesive Environmental, Social, and Governance (ESG) themes (Sultana and Zeya, 2025). This process utilizes K-means clustering on vector representations to group synonymous terms (e.g., "GHG," "CO2 footprints," and "carbon intensity"), reducing feature dimensions by up to 90% while maintaining informativeness (Sultana and Zeya, 2025; Fodeh et al., 2011).

Explainability is further operationalized through post-hoc tools:

- **SHAP and LIME:** Used for token- and phrase-level feature attribution to visualize which linguistic cues trigger greenwashing labels (Prajwalasimha et al., 2025; Černevičienė and Kabašinskas, 2024).
- **Sankey Diagrams:** Visualizing the flow of evidence from raw text into final thematic clusters (Yim et al., 2025).

4.4 Neurosymbolic Regulatory Reasoning

The framework integrates neural pattern recognition with symbolic logic to enforce regulatory consistency (Heever et al., 2024; Pakina and Pujari, 2024). This neurosymbolic engine embeds formal taxonomies (e.g., EU Taxonomy, CSRD) as logical constraints, ensuring that AI-generated scores do not violate established legal guardrails (Wu and Zhao, 2025; Pakina and Pujari, 2024). Differentiable fuzzy semantics are utilized to align neural outputs with symbolic representations, providing traceable reasoning chains that satisfy high-stakes accountability standards (Wu and Zhao, 2025; Peer and Stabinger, 2025).

4.5 Knowledge Graph-extended RAG (KG-RAG)

To mitigate hallucination risks, the study employs **KG-RAG**, which grounds Large Language Model (LLM) outputs in a structured, domain-specific Knowledge Graph (e.g., EmeraldGraph) (Kaoukis et al., 2025; Linders and Tomczak, 2025). The graph semantically links entities, risks, and regulatory requirements, allowing auditors to navigate the lineage of a specific metric from a high-level score down to sentence-level evidence (Nassar et al., 2025; Yim et al., 2025).

4.6 Scalability and Verification for SMEs

For resource-constrained Small and Medium-sized Enterprises (SMEs), the methodology utilizes a **”bridging strategy”** (Chou, 2024). Causal inference models are first trained on comprehensive data from large-cap firms and then adapted to the SME context through customized, materiality-adjusted indicators (Chou, 2024; Alkatheeri et al., 2023).

System robustness is maintained through the **Chain-of-Verification (CoVe)** protocol, which iteratively validates generated claims against retrieved evidence (Wu et al., 2025). This verification layer has been shown to reduce hallucination rates from 16% to 10%, ensuring that automated assessments do not facilitate ”automated greenwashing” (Wu et al., 2025; Peer and Stabinger, 2025).

4.7 Evaluation Metrics

The quantitative reliability of the framework will be assessed using the following metrics:

- **Quadratic Weighted Kappa (QWK):** Measuring agreement levels with expert human auditors, targeting a benchmark of 0.72–0.78 (Kumar and Boulanger, 2021).
- **Cohen’s Kappa:** Quantifying inter-rater agreement during Level-2 audits (Yim et al., 2025).
- **Constraint Violation Rate:** Measuring the frequency of logical contradictions within the neurosymbolic engine (Wu et al., 2025).

Layer	Technological Framework	Outcome
Intelligence	FinBERT + ABSA	Tone-aware aspect extraction (Huang et al., 2022)
Explanation	SHAP + LIME	Feature-level attribution (Prajwalasimha et al., 2025)
Reasoning	Neurosymbolic Engine	Taxonomy-aligned logic (Heever et al., 2024)
Verification	KG-RAG + CoVe	Grounded, hallucination-free verdicts (Wu et al., 2025)

Table 1: Methodological Layers and Technical Integration.

5 Contributions and Expected Outcomes

5.1 Scientific Contributions

This research is expected to advance the state of the art across multiple scientific dimensions spanning Natural Language Processing, Explainable AI, neurosymbolic reasoning, and sustainable finance analytics.

1. **Explainable Linguistic Detection of Impression Management.** The project will establish new methodological foundations for integrating explainable Aspect-Based Sentiment Analysis with regulatory-grade interpretability, enabling systematic identification of linguistic strategies associated with carbon-washing and strategic fluffing at sentence and aspect levels.
2. **Neurosymbolic ESG Reasoning Under Regulatory Constraints.** The research will contribute novel hybrid architectures that encode formal regulatory logic directly into neural learning pipelines, empirically quantifying the trade-offs between predictive accuracy, interpretability, and regulatory consistency.
3. **Justification-Centric Multimodal ESG Intelligence.** By extending Retrieval-Augmented Generation with Knowledge Graph grounding and multimodal evidence alignment, the project will advance explainable generation paradigms suitable for high-stakes verification tasks.
4. **Robustness and Self-Correction in Generative Systems.** The integration of Chain-of-Verification and compliance validation mechanisms will contribute new empirical insights into reducing hallucinations and error propagation in applied generative AI systems.
5. **Formalization of Auditability Metrics for AI Systems.** The research will operationalize auditability as a measurable construct, introducing evaluation protocols that quantify trace completeness, explanation fidelity, and regulatory alignment.

5.2 Technical Artifacts and Deliverables

Technical Artifacts encompass open-source explainable ABSA pipelines, regulatory ontologies and symbolic rule libraries, Knowledge Graph schemas and KG-RAG retrieval pipelines, benchmark datasets and annotated corpora, evaluation toolkits, and demonstration dashboards

5.3 Societal, Policy, and Industry Impact

Societal Policy Impact extends to investor protection, regulatory enablement under CSRD and EU Taxonomy, SME inclusion in sustainable finance, trustworthy AI deployment principles, and market integrity enhancement

5.4 Academic Dissemination and Knowledge Transfer

Expected dissemination channels include:

- Peer-reviewed journal publications in NLP, AI governance, and sustainable finance.
- Conference presentations (ACL, EMNLP, NeurIPS, AAAI, KDD, ICML workshops).
- Open datasets and benchmarks hosted on public repositories.
- Industry white papers and regulatory policy briefs.
- Open-source software releases with documentation and reproducibility packages.

6 Logistics, Timeline, Ethics, and Data Management

6.1 Project Timeline and Milestones

The doctoral project is structured over a four-year horizon, organized into progressive phases that support incremental validation and risk mitigation.

Period	Milestones
Year 1	Literature consolidation; dataset acquisition; regulatory ontology design; baseline ABSA and sentiment pipelines; preliminary explainability experiments; confirmation review.
Year 2	Explainable ABSA refinement; neurosymbolic rule encoding; KG schema design; pilot KG-RAG implementation; internal evaluations; first journal or conference submission.
Year 3	Multimodal integration; robustness and self-correction modules; large-scale benchmarking; audit simulation studies; industrial collaboration (if applicable); second publication cycle.
Year 4	System consolidation; longitudinal validation; thesis writing; defense preparation; technology transfer and open-source release.

6.2 Resource Requirements and Infrastructure

The project requires access to:

- GPU-enabled computing resources for model training and evaluation.
- Secure storage for document corpora and experimental artifacts.
- Version control and experiment tracking platforms.
- Cloud infrastructure for scalable experimentation (if required).
- Annotation tools for labeling and validation.

Existing institutional infrastructure and open-source tooling are expected to satisfy the majority of computational and software needs.

6.3 Ethical Considerations

The research primarily utilizes publicly available corporate disclosures, regulatory texts, and secondary datasets. No direct human subjects experimentation is planned. Nevertheless, several ethical dimensions will be addressed:

- **Data Privacy:** Compliance with GDPR and institutional data protection policies.
- **Algorithmic Bias:** Monitoring and mitigation of sectoral, geographic, or linguistic bias.
- **Transparency and Accountability:** Preservation of explanation fidelity and traceability.
- **Misuse Prevention:** Safeguards against dual-use deployment for deceptive reporting.
- **Human Oversight:** Integration of expert validation in high-stakes evaluation.

Formal ethical approval will be obtained if required by institutional policy.

6.4 Data Management and Reproducibility

The project will adhere to FAIR data principles (Findable, Accessible, Interoperable, Reusable).

- Versioned datasets and metadata documentation.
- Secure storage and access control.
- Open-source code repositories with reproducibility pipelines.
- Model checkpoints with documented training configurations.
- Long-term archival via institutional repositories.

Sensitive or licensed datasets will be handled in accordance with contractual restrictions.

6.5 Risk Management

- **Data Availability Risk:** Mitigated via diversified data sources and synthetic augmentation.
- **Computational Risk:** Modular scaling and cloud fallback.
- **Regulatory Change Risk:** Ontology versioning and rule modularity.
- **Model Reliability Risk:** Multi-layer verification and continuous evaluation.
- **Timeline Risk:** Phase-based milestones with contingency buffers.

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A Supplementary Materials

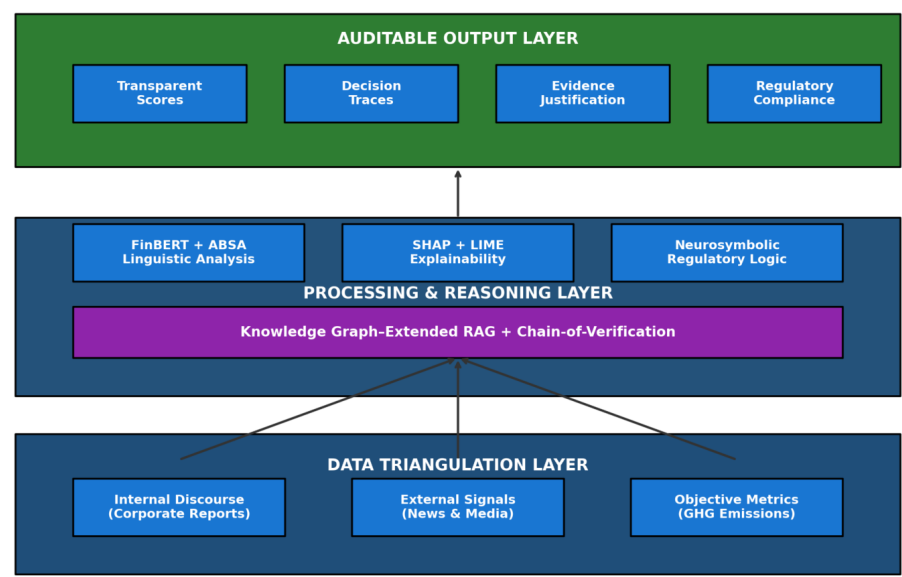


Figure 1: 3-layer architecture for Auditable AI framework

This diagram illustrates the three-layer architecture of your auditable AI framework. The Data Triangulation Layer demonstrates how your system integrates three distinct data streams to validate corporate ESG claims. Internal discourse from corporate reports is cross-referenced with external signals from news and media, alongside objective metrics such as greenhouse gas emissions. This triangulation approach is crucial for detecting the discourse-practice gap where corporate narratives are decoupled from substantive environmental actions.

The Processing Reasoning Layer shows the integration of FinBERT for linguistic analysis, SHAP and LIME for explainability, neurosymbolic regulatory logic, and Knowledge Graph-extended RAG with Chain-of-Verification. Finally, the Auditable Output Layer produces transparent scores, decision traces, evidence justifications, and regulatory compliance assessments that transform opaque black box ratings into auditable open box intelligence.

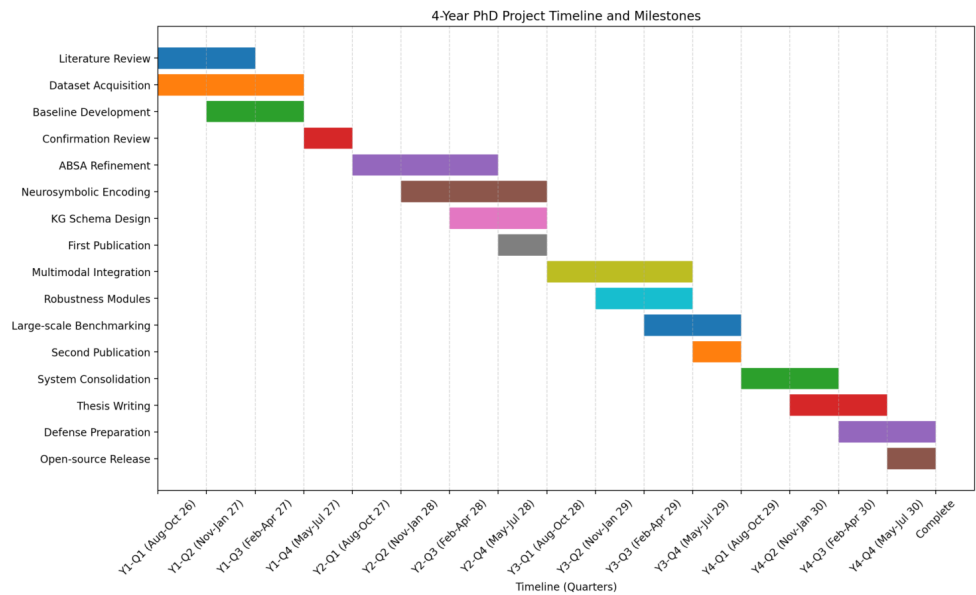


Figure 2: PhD Timeline

This Gantt chart presents the four-year doctoral timeline with progressive phases supporting incremental validation and risk mitigation. Year 1 focuses on literature consolidation, dataset acquisition, regulatory ontology design, baseline ABSA development, and preliminary explainability experiments culminating in a confirmation review.

Year 2 involves ABSA refinement, neurosymbolic rule encoding, KG schema design, pilot KG-RAG implementation, and the first publication cycle. Year 3 emphasizes multimodal integration, robustness modules, large-scale benchmarking, audit simulation studies, and industrial collaboration leading to a second publication. Year 4 concentrates on system consolidation, longitudinal validation, thesis writing, defense preparation, and technology transfer through open-source release

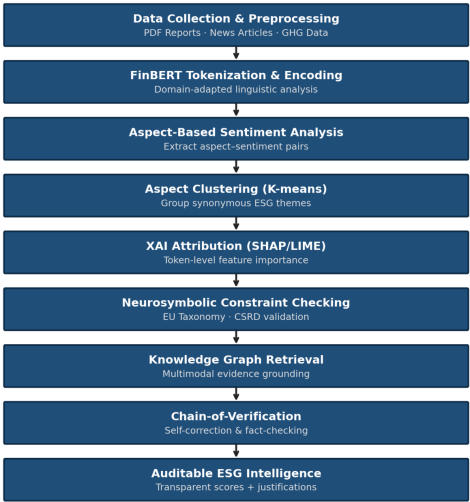


Figure 3: Methodology Pipeline

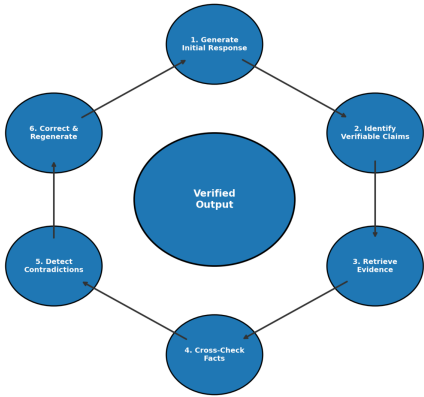


Figure 4: Chain-of-Verification self-correction pipeline

Methodology Pipeline. This comprehensive flowchart maps the complete methodology pipeline from raw data collection to auditable intelligence. The process begins with data collection and preprocessing of PDF reports, news articles, and GHG data, followed by FinBERT tokenization which captures domain-specific nuances that generic models frequently mislabel as neutral. Aspect-Based Sentiment Analysis extracts fine-grained aspect–sentiment pairs, followed by K-means clustering that aggregates data points into cohesive ESG themes, reducing feature dimensions by up to 90% while maintaining informativeness. The pipeline incorporates explainability through SHAP and LIME for token-level feature attribution, neurosymbolic constraint checking against EU Taxonomy and CSRD validation, knowledge graph retrieval for multimodal evidence grounding, and Chain-of-Verification for self-correction and fact-checking.

Chain-of-Verification (CoVe). This circular flow diagram illustrates the CoVe protocol that implements automated self-correction to reduce hallucination risk in high-stakes regulatory environments. The six-step iterative process begins with generating an initial response, followed by identifying verifiable claims. The system retrieves evidence from the knowledge graph, cross-checks facts, detects contradictions, and corrects and regenerates responses. This verification layer has demonstrated significant performance improvements, reducing hallucination rates from 16% to 10%. The protocol ensures factual integrity through continuous validation and supports auditable intelligence in regulatory deployment contexts.

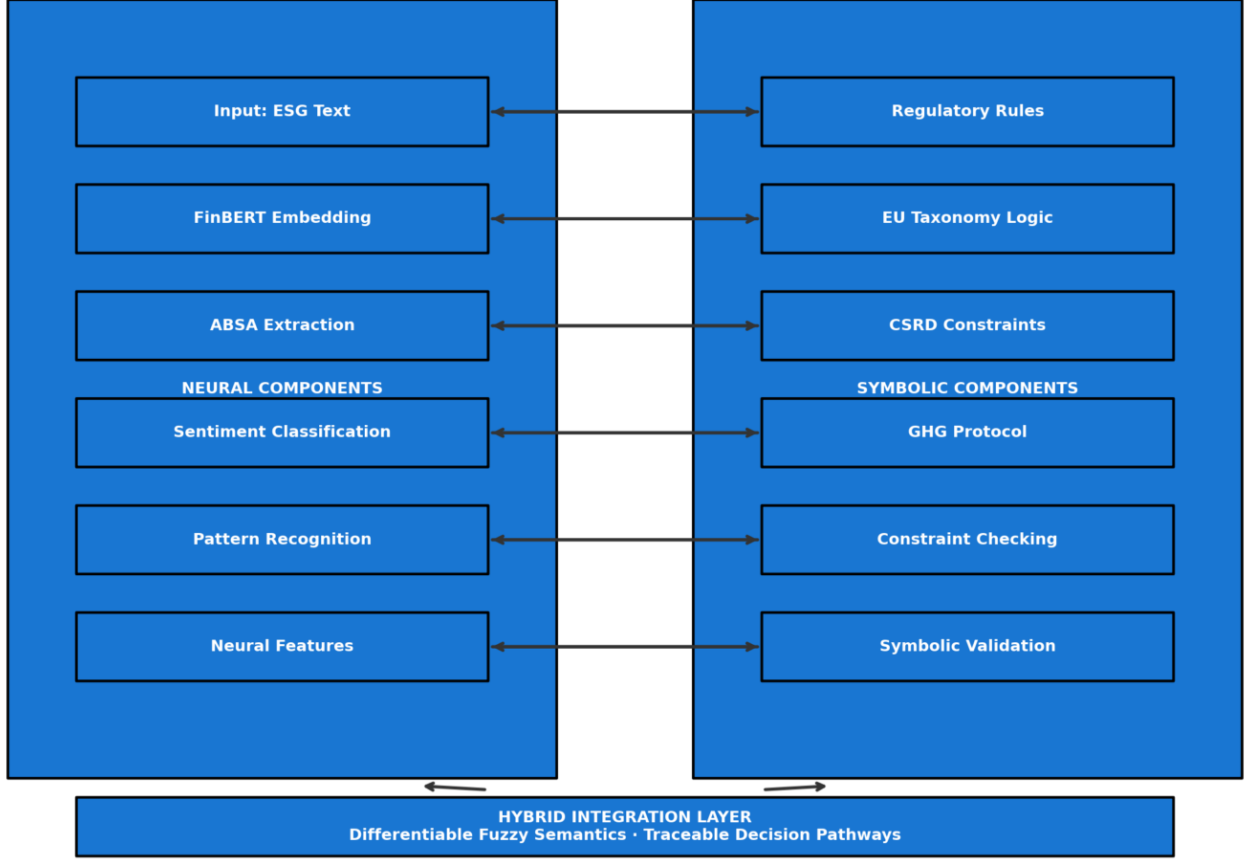


Figure 5: System architecture of the proposed framework

This diagram visualizes the hybrid neurosymbolic architecture that reconciles the trade-off between predictive performance and interpretability. The Neural Components side shows the progression from input ESG text through FinBERT embedding, ABSA extraction, sentiment classification, pattern recognition, to neural features. FinBERT’s domain-specific knowledge identifies approximately 18% more textual informativeness than generic architectures by capturing abnormal positive tone used to obfuscate poor sustainability performance.

The Symbolic Components side encompasses regulatory rules, EU Taxonomy logic, CSRD constraints, GHG Protocol, constraint checking, and symbolic validation. These components establish formal taxonomies as logical constraints, ensuring AI-generated scores do not violate established legal guardrails. The bidirectional arrows represent the continuous interaction between neural and symbolic processing, while the Hybrid Integration Layer employs differentiable fuzzy semantics to provide traceable decision pathways. This architecture addresses the critical gap where high-performing deep learning models lack the logical rigor of symbolic reasoning.